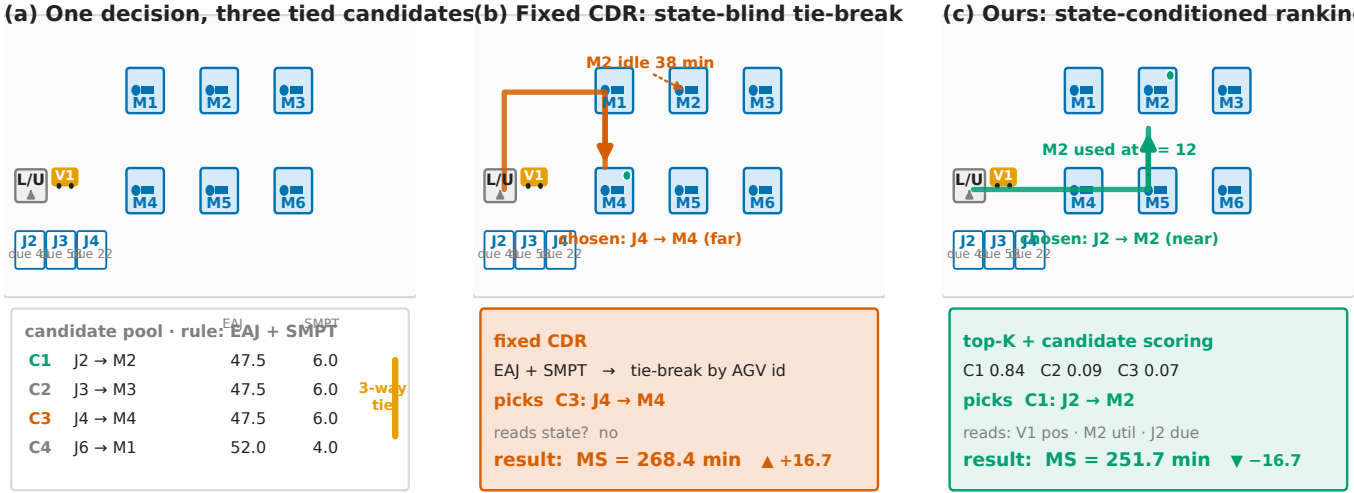

(a) One decision, three tied candidates
M1
M2
M3
M4
M5
M6
L/U
V1
J2
J3
J4
due 4
due 5
due 22
candidate pool · rule: EAJ + SMPT
EAJ
SMPT
C1 J2 → M2 47.5 6.0
C2 J3 → M3 47.5 6.0
C3 J4 → M4 47.5 6.0
C4 J6 → M1 52.0 4.0
3-way tie
(b) Fixed CDR: state-blind tie-break
M2 idle 38 min
chosen: J4 → M4 (far)
fixed CDR
EAJ + SMPT → tie-break by AGV id
picks C3: J4 → M4
reads state? no
result: MS = 268.4 min ▲ +16.7
(c) Ours: state-conditioned ranking
M2 used at = 12
chosen: J2 → M2 (near)
top-K + candidate scoring
C1 0.84 C2 0.09 C3 0.07
picks C1: J2 → M2
reads: V1 pos · M2 util · J2 due
result: MS = 251.7 min ▼ −16.7